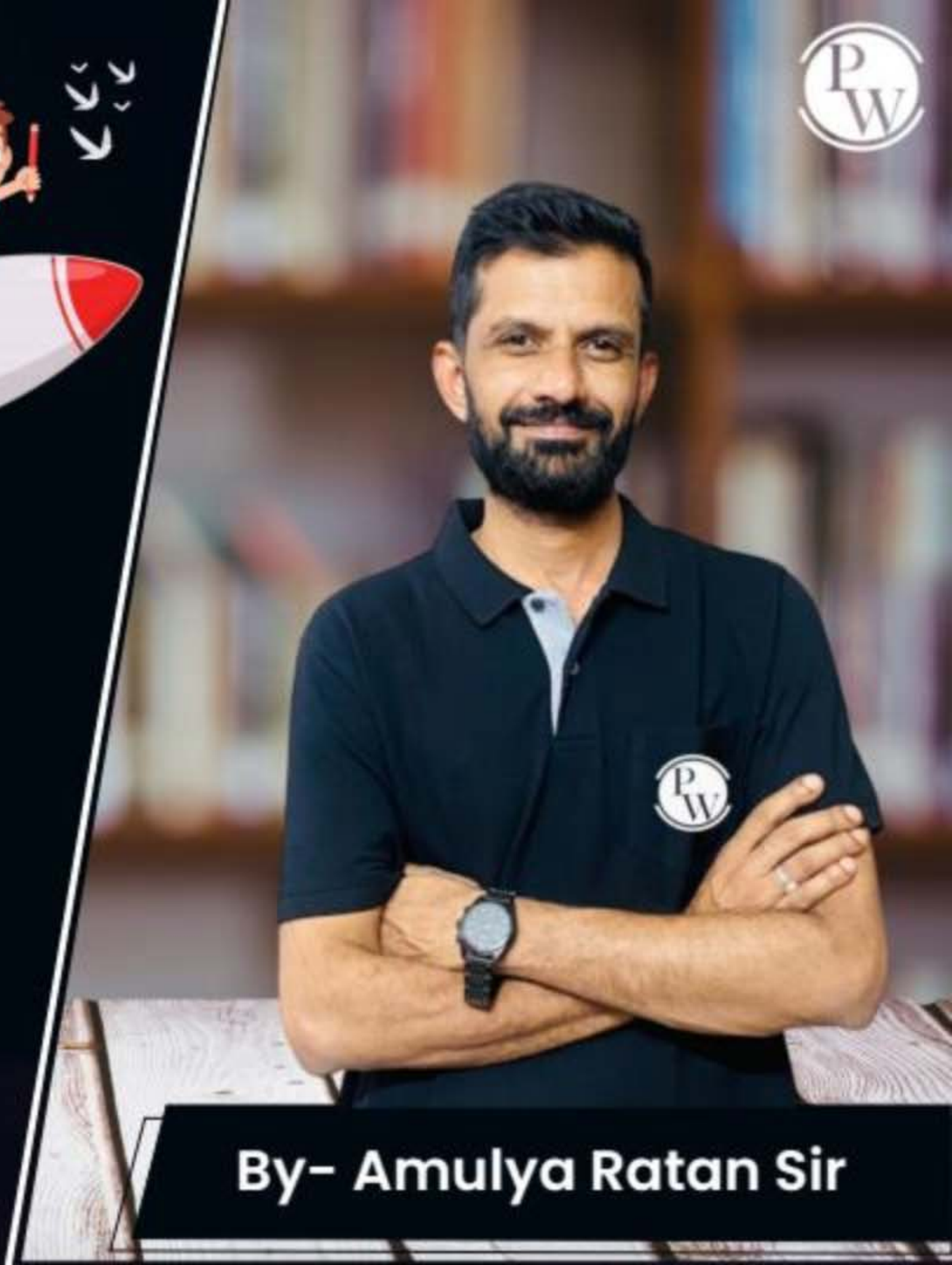
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 09

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Recap of Previous Lecture

Topic Number System

Topics to be Covered

Topic-1 More on Numbers

Topic-2 Counting Theory

[MCQ]

#Q. What would the unit digit in the answer of given expression:

1! + 2! + 3! +.....2025!

Assignment

1! → 1

2! → 2

3! → 6

4! → 24

5! → 120

3

6! → 120 × 6 = 720

7! → × 7 = 5040

[MCQ]

#Q. What would the remainder when the answer of given expression is divided by 3?

$$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2025! = \text{Answer}$$

Assignment

3

(A) 0

(B) 1

(C) 2

(D) Can't be determined

R = ?

[MCQ]

#Q. How many zeroes would be at the end in the answer of 50! ?

Assignment

$25 = 5 \times 5$

12 zeroes

$25 \times 2 = 50$

$50 \times 49 \times 48 \times 47 \times \ldots \ldots 2 \times 1$

OR

$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \ldots \ldots 49 \times 50$

1 – 10 → 2

11 – 20 → 2

21 – 30 → 3

31 – 40 → 2

41 – 50 → 3

12

A. 12

B. 15

C. 23

D. 24

$$\frac{50}{5} \rightarrow 10$$

$$\frac{10}{5} \rightarrow 2$$

$$\frac{2}{5} \rightarrow 0$$

$$\underline{\underline{12}}$$

[MCQ]

#Q. If 50! can be denoted maximum 7^x, then the value of x is?

8

$50! \Rightarrow 7$

$\frac{50}{7} \rightarrow 7$

$\frac{7}{7} \rightarrow 1$

$\frac{1}{7} \rightarrow 0$

8

1 – 10 → 1

11 – 20 → 1

21 – 30 → 2

31 – 40 → 1

41 – 50 → 3

8

A. 9

B. 8

C. 7

D. None of these

$48! \rightarrow 9^{11}$

3×3

$\frac{48}{3} \rightarrow 16$

$\frac{16}{3} \rightarrow 5$

$\frac{5}{3} \rightarrow 1$

$\frac{1}{3} \rightarrow 0$

$22 \rightarrow 3$

$48! \rightarrow 9^{11}$

$$48! \rightarrow \overset{\cancel{32}}{6}$$

$$2 \times 3$$

$$46 \qquad 22$$

$$\frac{48}{2} \rightarrow 24$$

$$\frac{24}{2} \rightarrow 12$$

$$\frac{12}{2} \rightarrow 6$$

$$\frac{6}{2} \rightarrow 3$$

$$\frac{3}{2} \rightarrow 1$$

$$\frac{1}{2} \rightarrow 0$$

$$50! \rightarrow \underline{70}^{8}$$

$$7 \times 5 \times 2$$

$$\underline{8} \quad \underline{12} \quad 47$$

$$\frac{50}{2} \rightarrow 25$$

$$\frac{25}{2} \rightarrow 12$$

$$\frac{12}{2} \rightarrow 6$$

$$\frac{6}{2} \rightarrow 3$$

$$\frac{3}{2} \rightarrow 1$$

$$\frac{1}{2} \rightarrow 0$$

$$\frac{50}{7} \rightarrow 7$$

$$\frac{7}{7} \rightarrow \underline{1}$$

$$8$$

$$\frac{50}{5} \rightarrow 10$$

$$\frac{10}{5} \rightarrow 2$$

$$\frac{2}{5} \rightarrow \underline{0}$$

65

$50! \Rightarrow \frac{12}{3 \times 2 \times 2}$

22

22 23

$\frac{50}{3} \Rightarrow 16$

$\frac{16}{3} \Rightarrow 5$

$\frac{5}{3} \Rightarrow 1$

$\frac{1}{3} \Rightarrow 0$

22

$\frac{50}{2} \Rightarrow 25$

$\frac{25}{2} \Rightarrow 12$

$\frac{12}{2} \Rightarrow 6$

$\frac{6}{2} \Rightarrow 3$

$\frac{3}{2} \Rightarrow 1$

$\frac{1}{2} \Rightarrow 0$

47

[MCQ]

#Q. Find the number of zeroes in following multiplication:

5 x 10 x 15 x 20 x 25..... x 50 ?

$\frac{10}{5} \to 2$

$\frac{2}{5} \to 0$

$= 5\times1\times5\times2\times5\times3\times5\times4 \_\_\_\_ 5\times10$

$= 5^{10} \times 10!$

$\frac{10}{2} \to 5$

$\frac{5}{2} \to 2$

$\frac{2}{2} \to 1$

$\frac{1}{2} \to 0$

5 → 12

2 → 8

A. 48

B. 12

C. 24

D. 8

[MCQ]

#Q. How many squares are there in a chessboard?

Size	No.
1x1	8x8 = 64
2x2	7x7 = 49
3x3	6x6 = 36
4x4	5x5 = 25
5x5	4x4 = 16
6x6	3x3 = 9
7x7	2x2 = 4
8x8	1x1 = 1

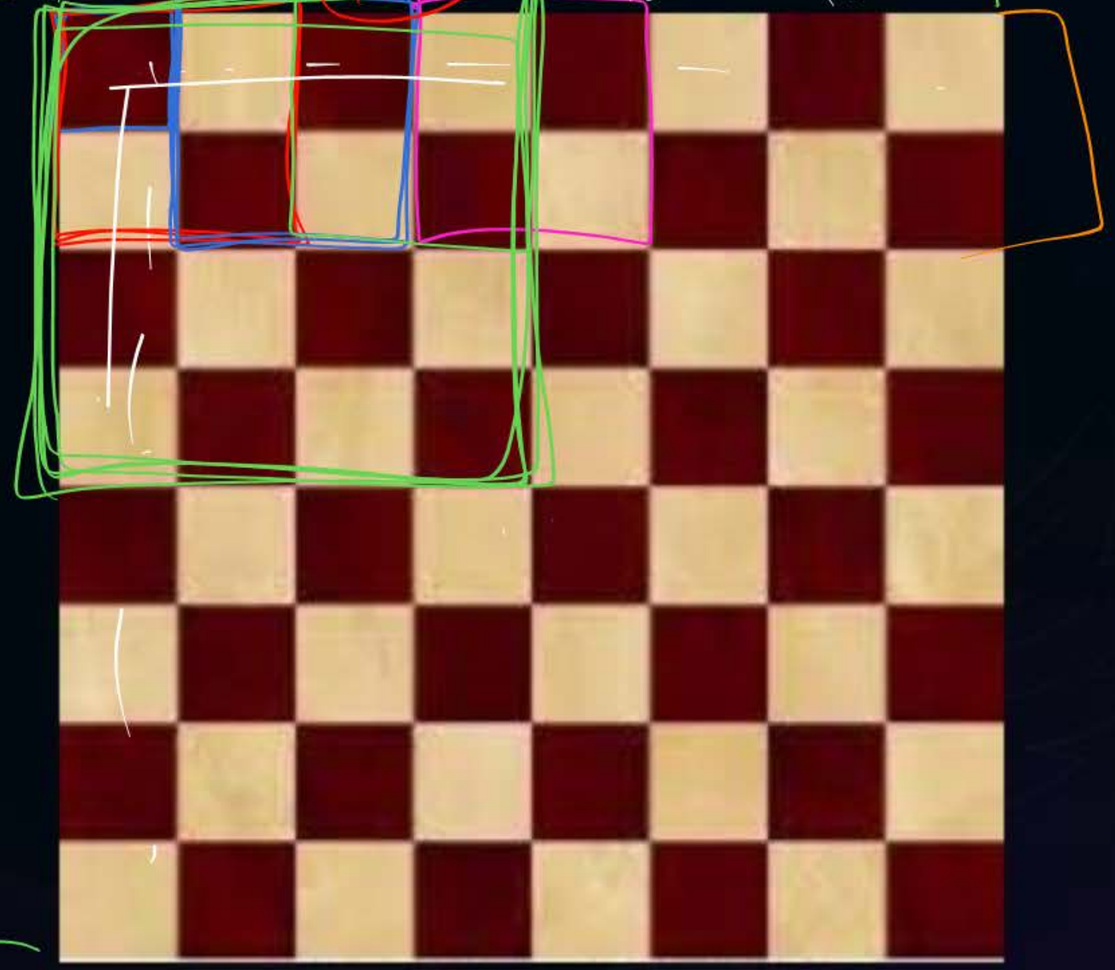
204

Size: 1x1, 2x2, 3x3, 4x4

4x4 = 16
3x3 = 9
2x2 = 4
1x1 = 1

30

8



[MCQ]

#Q. What would be the unit digit in the answer of: $2362^{911} + 1418^{216}$

$\rightarrow 4$

$8 + 6 = 14$

64×8

$4)911(227$

8

11

8

31

28

$R = 3$

$2^1 \rightarrow 2 \quad R=1$

$2^2 \rightarrow 4 \quad R=2$

$2^3 \rightarrow 8 \quad R=3$

$2^4 \rightarrow 6 \quad R=0$

$2^5 \rightarrow 2$

$2^6 \rightarrow 4$

$8^1 \rightarrow 8$

$8^2 \rightarrow 4$

$8^3 \rightarrow 2$

$8^4 \rightarrow 6 \quad R=0$

$8^5 \rightarrow 8$

[MCQ]

#Q. What would be the unit digit in the answer of: $2913^{1902} + 1647^{460}$?

$9 + 1 = 10$

$3^1 \to 3$
$3^2 \to 9$
$3^3 \to 7$
$3^4 \to 1$
$3^5 \to 3$

$R = 2$

$7^1 \to 7$
$7^2 \to 9$
$7^3 \to 3$
$7^4 \to 1$
$7^5 \to 7$

$R = 0$

A. 0

B. 2

C. 3

D. 5

[MCQ]

#Q. What would be the unit digit in the answer of: $9326^{397} + 1475^{363}$

$6 + 5 = 11$

0 1 5 6 → Same no.

4 → odd: 4, even: 6

9 → odd: 9, even: 1

2 3 7 8

1: 4 → 4

2: 4 → 6

3: 4 → 4

[MCQ]

#Q. What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1 ?

A. 9997

B. 9793

C. 9895

D. 9487

9792+1

=9793

612

1224

1836

9180 (612×15)

9792 (612×16)

2	6, 9, 12, 17
2	3, 9, 6, 17
3	3, 9, 3, 17
	1, 3, 1, 17

36×17 = 612

613

Divisibility Rules:

$16 \rightarrow 2^4$

$32 \rightarrow 2^5$

$\boxed{46983}$ $= 31$

- 2: → Any given no, if last digit is 0 OR multiple of 2
- 3: → Sum of the digits of a given number should be divisible by 3
- 4: → $4 \rightarrow 2^2$ $8 \rightarrow 2^3$

last two digit 00 OR multiple of 4

last three digit 000 OR multiple of 8

$46983 = 30$

Divisibility Rules:

00 OR multiple of 25 → last 2 digit

25 → 5

- 5: → last digit of a given number should be 0 OR Multiple of 5
- 6: = 2 × 3
- 7:

$$\boxed{48937\,7}$$

$48937 \sim 14$

$= 4892\,3$

$4892 \sim 6$

$= 4886$

$488 - 12$

$= 476$

$47 \sim 12$

$= 35$

Twin Prime & Co-Prime Numbers:

11, 13 ✓
3, 5 ✓
17, 19 ✓
5, 7 ✓
29, 31 ✓

21 → 1, 3, 7, 21
22 → 1, 2, 11, 22

9 → 1, 3, 9
16 → 1, 2, 4, 16

1. Any two distinct prime no
2. Any two consecutive number

Divisibility Rules:

$100 \rightarrow 10^{2}$

- 8: $= 2^{3}$ Last three digit 000 OR multiple of 8
- 9: → Sum of the digits of a given no should be divisible by 9.
- 10: →

Divisibility Rules:

- 11:
- 12: = 3 × 4
- 14: = 2 × 7

2 → 1, 2
6 → 1, 2, 3, 6

2 + 8 + 1 = 11

129831

1 + 9 + 3 + 1 = 14

3

OR

Divisibility Rules:

- 15: → 3×5
- 16: → 2^4 → last 4 digits 0000 or Multiple of 16
- 18: → 2×9

Divisibility Rules:

- 20: → 4×5
- 40: → 5×8
- 80: → 16×5

[MCQ]

#Q. Which largest number of 5 digits is divisible by 99 ?

Assignment

A 99999

B 99981

C 99909

D 99990

[MCQ]

#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

Assignment

A. 216

B. 1728

C. 512

D. 360

[MCQ]

#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

Assignment

A. 13

B. 14

C. 17

D. 16

[MCQ]

#Q. Find the largest four digit number that is exactly divisible by 88.

Assignment

A. 9844

B. 9768

C. 8894

D. 9944

2 mins Summary

Topic ~~Counting~~ Theory

Numbers

THANK - YOU